YOOZOO Games: Offering glitchless gaming experiences to players around the world with an extensive global network

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around the world
Games leverages
extensive global r
Google Cloud to r

About YOOZOO Games

Founded in 2009, YOOZOO Games is headquartered in Shanghai and has branch offices in more than 10 countries, including Germany, Singapore, Japan, South Korea and India. Since June 2014, the company has been listed on the main board of China's stock market. YOOZOO Games specializes in globalized game R&D and publishing and has released several well-known games like the Junior Three Kingdom series, the League of Angels series, Daomu Biji, Game of Thrones: Episode 1, and Legend of Sanctuary. Having more than 1,000

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partners overseas, the company offers games to nearly 1 billion users in more than 200 countries and regions across Europe, North America, Middle East, Asia and South America.

Industries: Telecommunications, Media & Entertainment, Gaming

Location: China

Google Cloud results

- Enables stable data transferring across servers around the world with global VPC
- Reduces operational costs by 40% with custom VMs on Compute Engine
- Clears connection disruptions with Cloud Load Balancing

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(https://cloud.google.com/automl/), BigQuery

Google Cloud (https://cloud.google.com/)

Cloud AutoML (https://cloud.google.com/automl/)

BigQuery (https://cloud.google.com/bigquery/)

Google Cloud Armor (https://cloud.google.com/armor/)

Cloud CDN (https://cloud.google.com/cdn/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Compute Engine (https://cloud.google.com/compute/)

Virtual Private Cloud (https://cloud.google.com/vpc/)

- Increases ad clickthrough rate by 300% through precision targeting supported by AutoML

Thanks to the higher performance of smartphones and the growing speed of wireless internet, mobile games have become increasingly popular and diverse. Gamers can now enjoy a wide variety of video games anytime and anywhere on their mobile devices, including those that were traditionally only available on PCs and game consoles, such as player-versus-player (PvP) combat games.

Dedicated to offering the best-possible gaming experiences across various game types, YOOZOO Games (https://www.yoozoo.com/) is always looking for ways to expand its game portfolio. The company has released more than 30 games and currently has around 10 million users around the world. In late 2021, the company launched Metal Revolution (https://play.google.com/store/apps/details?id=com.gtarcade.next.mrglo&hl=en&gl=US), its first mobile PvP combat game, where gamers can enjoy the thrill of fighting against other players with their best tricks in real time in front of an audience. The game has received a high rating of

4.5 stars on Google Play, with some users hailing it as "probably the best fighting game on mobile."

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> *—Songtao Lu, Head of Operations, YOOZOO*

According to Songtao Lu, head of operations at YOOZOO Games, the release of Metal Revolution was an important milestone for the company because it is not easy to establish a well-connected cross-region server network that can support real-time combats between players around the world. "High connectivity is key to offering excellent gaming experiences in PvP combat games. Even slight latency can lead to the loss of one side, which would make gamers feel that the battle is unfair. Also, the audience of the fights would lose interest if they can't follow the latest moves due to slow or unstable networks," he adds.

Prior to the release of Metal Revolution, YOOZOO Games tested the game on a few public cloud platforms in North and South America. However, the network connectivity was not ideal, as high latency and frequent fluctuations significantly undermined the user experience. To enhance the quality of its combat game, YOOZOO Games migrated Metal Revolution in mid-2021 to Google Cloud (https://cloud.google.com/).

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Shortening in-game latency and supporting stable connection with VPC

YOOZOO Games now employs global Virtual Private Cloud (https://cloud.google.com/vpc) (VPC) to connect its servers of Metal Revolution in North and South America, and Cloud CDN (https://cloud.google.com/cdn) to support content delivery. Since VPC provides one single network across multiple regions, which helps enhance cross-region connectivity, and Cloud CDN enables faster content delivery by sending data from the server that is nearest to end users, YOOZOO Games is able to greatly increase the connection quality of its game. Before moving to Google Cloud, Metal Revolution's gamers experienced latency of more than 130 microseconds, with great instability. After the migration, the connectivity has been greatly improved with 80% of the users of Metal Revolution enjoying latency of less than 80 microseconds.

The fact that VPC provides one single global network means that the YOOZOO Games team doesn't have to find ways to connect its networks in different regions, which has also helped save costs and improve network stability. Before the migration, the company needed to connect the servers of Metal Revolution in North and South America with the database of the game hosted in Singapore by itself. Since the networks in different regions were separated, the connection was not always stable, which frequently caused delays or data loss. The company sometimes also needed to purchase extra bandwidth to ensure smooth data transferring. With global VPC coverage, all these cross-region connection issues have been solved effectively.

"With the single global network supported by VPC, the servers in different regions are connected through intranets, which guarantees low latency and high stability. As a result, our users can fully enjoy the excitement of real-time battles, and the number of user reports about in-game latency has greatly reduced," notes Chao Chen, operations architect at YOOZOO Games.

"The cloud platforms that we previously used in Taiwan only provide VMs with fixed ratios of CPUs and memory. As our game servers are mostly memory intensive, we ended up buying more CPUs than we needed. Compute Engine has helped us save operational costs by allowing us to purchase the exact computing resources required."

—Chao Chen, Operations Architect, YOOZOO Games

Saving operational costs with custom VMs on Compute Engine

High connectivity is not the only benefit that YOOZOO Games has gained from implementing Google Cloud infrastructure. To host its game servers, the company leverages Compute Engine (https://cloud.google.com/compute), which provides custom virtual machines (VMs). The flexibility of choosing VM specifications according to actual needs enables YOOZOO Games to use VMs with the lowest price-performance ratio. This advantage also prompted the company to move Junior Three Kingdom Zero and Junior Three Kingdom II, its two most popular card games in Taiwan, from a cloud platform that only offers fixed VM specifications to Google Cloud. After the migration, YOOZOO Games has cut down the operational costs of these two games by 40%.

Junior Three Kingdom Zero

Junior Three Kingdom 2

"The cloud platforms that we previously used in Taiwan only provide VMs with fixed ratios of CPUs and memory. As our game servers are mostly memory intensive, we ended up buying more CPUs than we needed," explains Chen. "Compute Engine has helped us save operational costs by allowing us to purchase the exact computing resources required."

Chen also shares that the CLI (https://cloud.google.com/sdk/gcloud) and API tools of Google Cloud have simplified the process of adjusting VM specifications. With these tools, YOOZOO Games' maintenance programmers are able to change the amount of CPUs and memory of the VMs on Compute Engine according to the needs in real time, instead of waiting for the DevOps team's decision to add or remove certain VMs.

Enhancing stability and security with Cloud Load Balancing and Cloud Armor

When it comes to enhancing its service quality, YOOZOO Games uses Cloud Load Balancing (https://cloud.google.com/load-balancing) to distribute incoming traffic among servers and Cloud Armor (https://cloud.google.com/armor) to prevent DDoS attacks. When a user tries to access games on Google Cloud, the request is first processed by Cloud Load Balancing and filtered through Cloud Armor. When traffic is high, Cloud Load Balancing can evenly allocate the traffic flow to different servers, which gives the YOOZOO Games team enough time to add sufficient computing resources and prevent its systems from crashing. On top of that, Cloud Load Balancing automatically performs health checks on server nodes and removes the malfunctioning ones, which helps ensure high stability and availability of YOOZOO Games' servers.

"We once tested Metal Revolution without using Cloud Load Balancing and found that 20% of users experienced connection disruptions, because our servers were overloaded or unresponsive," recalls Chen. "With Cloud Load Balancing, we've never encountered any server availability issues even under high traffic."

Supporting cost-effective advertising through AutoML

On the marketing side, YOOZOO Games wanted to further improve the profitability of its investments in advertising on Google Ads by leveraging the latest machine learning technology. In late 2021, Google gTech gPS, LCS and Cloud team formed one team and introduced the MOCHA solution, which offers end-to-end machine learning pipelines for advertising, to YOOZOO. With the assistance from Google, YOOZOO Games started uploading its user behavior data to BigQuery (https://cloud.google.com/bigquery) and employing AutoML (https://cloud.google.com/automl) to predict users' inclination to in-game purchase.

With the cutting-edge machine learning capability provided by AutoML, YOOZOO Games is able to identify potential and whale payers from newly installed users, which guides Google Ads for accurate user targeting thus highly optimized user acquisition. As of May 2022, the company successfully increased the ad clickthrough rate by 44.6% while reducing the CPI (cost per install) by 13%. With such successful results and Google One support, Yoozoo Games is more confident in investing more on building a powerful and automatic machine learning pipeline on top of Google Cloud to unlock the value behind game data.

"Google Cloud has helped us greatly in improving gaming experiences by supporting high connectivity across different regions and providing a reliable cloud infrastructure. Seeing the results so far, we're confident that we can offer even more exciting game features to players around the world in the future."

—Songtao Lu, Head of Operations, YOOZOO Games

Delivering higher game quality across the world

While developing new game products, YOOZOO Games continues to improve the gaming experiences of its existing portfolio. It plans to bring more games to Google Cloud for better connectivity and replace its self-built game database with Cloud SQL (https://cloud.google.com/sql), which can provide 99.95% availability SLA. The company will also expand its use of Cloud Load Balancing to manage traffic distribution between its game servers, which will save the cost of developing network gateways by itself and further enhance server stability.

"Google Cloud has helped us greatly in improving gaming experiences by supporting high connectivity across different regions and providing a reliable cloud infrastructure," says Lu. "Seeing the results so far, we're confident that we can offer even more exciting game features to players around the world in the future."